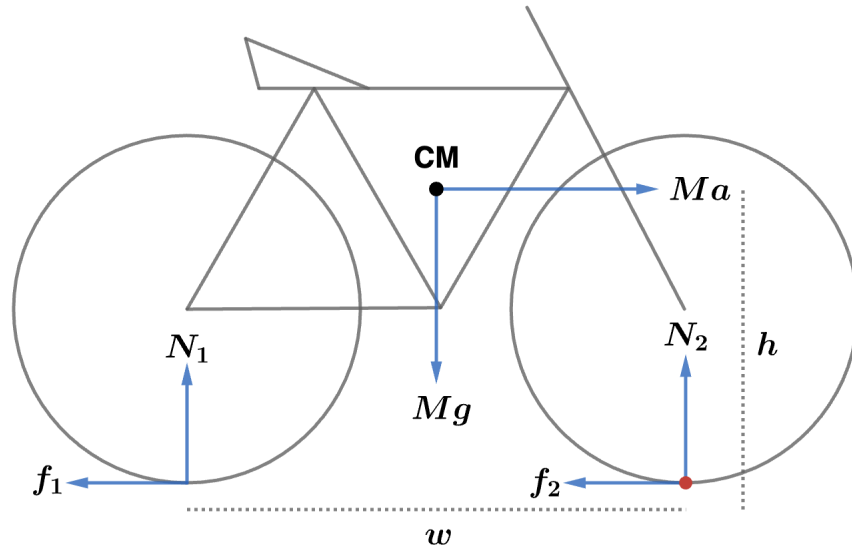


2007 F=ma Exam: Problem 29

Kevin S. Huang



We go to the accelerating frame to reduce the setup to a statics problem, introducing the fictitious inertial force $-M\vec{a}$ on the bicycle's CM. Choosing the contact point of the front wheel with the ground as the pivot point, we can balance torques:

$$Mah + N_1 w = Mg \left(\frac{w}{2} \right)$$

$$Mah = \left(\frac{Mg}{2} - N_1 \right) w \leq \frac{Mgw}{2}$$

using the fact that the normal force $N_1 \geq 0$. Then

$$a \leq \frac{gw}{2h}$$

so the answer is B.